T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

R_W	$16\pi^2\beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \to g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

Column	Representative term structure
T1	$\frac{43{g_1}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43{g_1}^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17{g_2}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17{g_2}^3}{6}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$
T2	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$
T3	$-8{g_3}^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$
T3	$-8{g_3}^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

$\beta_{y_1}^{(1)}$

Quantity: y_1

$$16\pi^2\,\mu\frac{dy_1}{d\mu} = \beta_{y_1}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$
T2	$+T_\nu^{(1)}\,(y_1)_{ij}$
T3	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij} + \left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$	$+T_\nu^{(1)}\,(y_1)_{ij}$	$+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij} + \left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$

Quantity: y_1

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{9}{4}{g_2}^2 (y_1)_{ij}$
T2	$-\frac{3}{4}{g_1}^2 (y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}{g_2}^2 (y_1)_{ij}$	$-\frac{3}{4}{g_1}^2 (y_1)_{ij}$

$\beta_{y_2}^{(1)}$

Quantity: y_2

$$16\pi^2\,\mu\frac{dy_2}{d\mu} = \beta_{y_2}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$
T2	$+T_\nu^{(2)}\,(y_2)_{ij}$
T3	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)(y_1)_{ij}$	$+T_\nu^{(2)}\,(y_2)_{ij}$	$+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$

Quantity: y_2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{9}{4}{g_2}^2 (y_2)_{ij}$
T2	$-\frac{3}{4}{g_1}^2 (y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}{g_2}^2 (y_2)_{ij}$	$-\frac{3}{4}{g_1}^2 (y_2)_{ij}$

$\beta^{(1)}_{M_F}$

Quantity: M_F

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu} = \beta^{(1)}_{M_F}$$

Terms 1 of 2

Column	Representative term structure
T1	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$
T2	$+\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
T3	$+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$	$+\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$

Quantity: M_F

Terms 2 of 2

Column	Representative term structure
T1	$+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$
T2	$-8\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\Big)$
T3	$-8\,\mathrm{Tr}\Big(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	$-8\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\Big)$	$-8\,\mathrm{Tr}\Big(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\Big)$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{9{g_2}^2 m_1^2}{2}$
T2	$-\frac{3{g_1}^2 m_1^2}{2}$
T3	$+4\lambda_{12}^{(1)} m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	$+4\lambda_{12}^{(1)} m_2^2$

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$
T2	$-8\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\Big)$
T3	$-8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	$-8\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\Big)$	$-8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{9{g_2}^2 m_2^2}{2}$
T2	$-\frac{3{g_1}^2 m_2^2}{2}$
T3	$+4\lambda_{12}^{(1)} m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$	$+4\lambda_{12}^{(1)} m_1^2$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Quantity: λ_1

Terms 2 of 6

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$
T2	$+8\lambda_7\,(\lambda_7)^*$
T3	$+4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+8\lambda_7\,(\lambda_7)^*$	$+4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: λ_1

Terms 3 of 6

Column	Representative term structure
T1	$+8\lambda_8 \left(\lambda_8\right)^*$
T2	$+4\lambda_5 \left(\lambda_5\right)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_5 \left(\lambda_5\right)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: λ_1

Terms 4 of 6

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$
T2	$-9{g_2}^2\lambda_1$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$	$+\frac{3{g_1}^4}{4}$

Quantity: λ_1

Terms 5 of 6

Column	Representative term structure
T1	$+12{\lambda_1}^2$
T2	$+4{\lambda_{HS_1}^{(1)}}^2$
T3	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Quantity: λ_1

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)\ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1)\ 2}$	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

Quantity: $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Terms 1 of 5

Column	Representative term structure
T1	$4\lambda^{(1)}_{S_1} T^{(1)}_{\nu}$
T2	$-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3	$+8\lambda_7\,(\lambda_7)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda^{(1)}_{S_1} T^{(1)}_{\nu}$	$-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	$+8\lambda_7\,(\lambda_7)^*$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+8\lambda_9 \left(\lambda_9\right)^*$
T2	$+24\lambda_{11} \left(\lambda_{11}\right)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_9 \left(\lambda_9\right)^*$	$+24\lambda_{11} \left(\lambda_{11}\right)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T2	$-9{g_2}^2\lambda_{S_1}^{(1)}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4{\lambda_{12}^{(1)}}^2$
T2	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
T3	$+4{\lambda_{HS_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda_{12}^{(1)}}^2$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+4{\lambda_{HS_1}^{(1)}}^2$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
T2	$+12{\lambda_{S_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$	$+12{\lambda_{S_1}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 5

Column	Representative term structure
T1	$4\lambda^{(1)}_{S_2} T^{(2)}_{\nu}$
T2	$-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$+8\lambda_8\,(\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda^{(1)}_{S_2} T^{(2)}_{\nu}$	$-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$+8\lambda_8\,(\lambda_8)^*$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+8\lambda_9 \left(\lambda_9\right)^*$
T2	$+24\lambda_{10} \left(\lambda_{10}\right)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_9 \left(\lambda_9\right)^*$	$+24\lambda_{10} \left(\lambda_{10}\right)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T2	$-9{g_2}^2\lambda_{S_2}^{(1)}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_2}^{(1)}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4{\lambda_{12}^{(1)}}^2$
T2	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$
T3	$+4{\lambda_{HS_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda_{12}^{(1)}}^2$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$	$+4{\lambda_{HS_2}^{(1)}}^2$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
T2	$+12{\lambda_{S_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$	$+12{\lambda_{S_2}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+24\lambda_7\ (\lambda_7)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+24\lambda_7\ (\lambda_7)^*$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 2 of 5

Column	Representative term structure
T1	$+3\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T2	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^*)$
T3	$+6\lambda_5\left(\lambda_5\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^*)$	$+6\lambda_5\left(\lambda_5\right)^*$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 4 of 5

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{(1)}_{HS_1}$
T2	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$
T3	$+6\lambda_1\lambda^{(1)}_{HS_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda^{(1)}_{HS_1}$	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	$+6\lambda_1\lambda^{(1)}_{HS_1}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 5 of 5

Column	Representative term structure
T1	$+4{\lambda^{(1)}_{HS_1}}^{2}$
T2	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$
T3	$+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda^{(1)}_{HS_1}}^{2}$	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$+\frac{3{\lambda^{(A)}_{HS_1}}^{2}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$+3\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$+3\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{10})^*)$
T2	$+24\lambda_8\,(\lambda_8)^*$
T3	$+6\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{10})^*)$	$+24\lambda_8\,(\lambda_8)^*$	$+6\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$-9{g_2}^2\lambda^{(1)}_{HS_2}$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{g_1}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 4 of 5

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{(1)}_{HS_2}$
T2	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$
T3	$+6\lambda_1\lambda^{(1)}_{HS_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda^{(1)}_{HS_2}$	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	$+6\lambda_1\lambda^{(1)}_{HS_2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+4{\lambda_{HS_2}^{(1)}}^2$
T2	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda_{HS_2}^{(1)}}^2$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda^{(1)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$
T2	$-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$-4\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(1)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$	$-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-4\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 6

Column	Representative term structure
T1	$+2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T2	$+24\lambda_9\,(\lambda_9)^*$
T3	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$+24\lambda_9\,(\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 6

Column	Representative term structure
T1	$+3\lambda_5 \left(\lambda_5\right)^*$
T2	$+\frac{9{g_2}^4}{4}$
T3	$-9{g_2}^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_5 \left(\lambda_5\right)^*$	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 6

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$-3{g_1}^2\lambda_{12}^{(1)}$
T3	$+4{\lambda_{12}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 6

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta^{(1)}_{\lambda_5}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$
T2	$-\frac{2}{3}\,(\lambda_7)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$+2\,(\lambda_8)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_5\left(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right)$	$-\frac{2}{3}\,(\lambda_7)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$	$+2\,(\lambda_8)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$-9{g_2}^2\lambda_5$
T2	$-3{g_1}^2\lambda_5$
T3	$+2\lambda_{12}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_5$	$-3{g_1}^2\lambda_5$	$+2\lambda_{12}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_1\lambda_5$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta^{(1)}_{\lambda^{(A)}_{12}}$

Quantity: $\lambda^{(A)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(A)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$
T2	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3	$-64\lambda_9\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(A)}_{12}(2T^{(1)}_{\nu} + 2T^{(2)}_{\nu})$	$-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$	$-64\lambda_9\,(\lambda_9)^*$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$-4\lambda_{10}(5\,(\lambda_{10})^{*}+(\lambda_{11})^{*})$
T2	$-4\lambda_{11}((\lambda_{10})^{*}+5\,(\lambda_{11})^{*})$
T3	$+\frac{4}{3}\lambda_{5}\,(\lambda_{5})^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{10}(5\,(\lambda_{10})^{*}+(\lambda_{11})^{*})$	$-4\lambda_{11}((\lambda_{10})^{*}+5\,(\lambda_{11})^{*})$	$+\frac{4}{3}\lambda_{5}\,(\lambda_{5})^{*}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{12}^{(A)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$-64\lambda_7\,(\lambda_7)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-64\lambda_7\,(\lambda_7)^*$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 2 of 4

Column	Representative term structure
T1	$+4\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T2	$+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$
T3	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_1\lambda^{(A)}_{HS_1}$
T2	$+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$
T3	$+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda^{(A)}_{HS_1}$	$+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$	$+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3	$-4\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	$-4\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 2 of 4

Column	Representative term structure
T1	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$
T2	$+64\lambda_8\,(\lambda_8)^*$
T3	$+16\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$	$+64\lambda_8\,(\lambda_8)^*$	$+16\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$
T2	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_1\lambda^{(A)}_{HS_2}$
T2	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$
T3	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda^{(A)}_{HS_2}$	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

$\beta^{(1)}_{\lambda_7}$

Quantity: λ_7

$$16\pi^2\,\mu\frac{d\lambda_7}{d\mu} = \beta^{(1)}_{\lambda_7}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-\frac{1}{3}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$-9{g_2}^2\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-\frac{1}{3}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$	$-9{g_2}^2\lambda_7$

Quantity: λ_7

Terms 2 of 3

Column	Representative term structure
T1	$-3{g_1}^2\lambda_7$
T2	$+2\lambda_1\lambda_7$
T3	$+8\lambda^{(1)}_{HS_1}\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda_7$	$+2\lambda_1\lambda_7$	$+8\lambda^{(1)}_{HS_1}\lambda_7$

Quantity: λ_7

Terms 3 of 3

Column	Representative term structure
T1	$-4\lambda_{HS_1}^{(A)}\,\lambda_7$
T2	$+2\lambda_7\lambda_{S_1}^{(1)}$
T3	$+4\lambda_8\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{HS_1}^{(A)}\,\lambda_7$	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$

$\beta^{(1)}_{\lambda_{12}}$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu} = \beta^{(1)}_{\lambda_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+6\lambda_9\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T3	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_9\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure
T1	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T3	$+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Quantity: λ_{12}

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$
T2	$+\lambda_{HS_1}^{(A)}\left(\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{3\lambda_{11}}{4}\right)$
T3	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{3\lambda_{11}}{4}\right)$	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$

Quantity: λ_{12}

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$

$\beta^{(1)}_{\lambda_8}$

Quantity: λ_8

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu} = \beta^{(1)}_{\lambda_8}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$
T3	$+4\lambda_7\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$	$+4\lambda_7\,(\lambda_9)^*$

Quantity: λ_8

Terms 2 of 3

Column	Representative term structure
T1	$-9{g_2}^2\lambda_8$
T2	$-3{g_1}^2\lambda_8$
T3	$+2\lambda_1\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_8$	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$

Quantity: λ_8

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda^{(1)}_{HS_2}\lambda_8$
T2	$+4\lambda^{(A)}_{HS_2}\lambda_8$
T3	$+2\lambda_8\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda^{(1)}_{HS_2}\lambda_8$	$+4\lambda^{(A)}_{HS_2}\lambda_8$	$+2\lambda_8\lambda^{(1)}_{S_2}$

$\beta^{(1)}_{\lambda_{11}}$

Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu} = \beta^{(1)}_{\lambda_{11}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$
T2	$+4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
T3	$-9{g_2}^2\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$	$+4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$

Quantity: λ_{11}

Terms 2 of 3

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{11}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda_{11}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2}\right)$

Quantity: λ_{11}

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda^{(1)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$+\frac{\lambda^{(A)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$
T3	$+12\lambda^{(1)}_{S_1}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda^{(1)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\frac{\lambda^{(A)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$	$+12\lambda^{(1)}_{S_1}\lambda_{11}$

$\beta^{(1)}_{\lambda_9}$

Quantity: λ_9

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu} = \beta^{(1)}_{\lambda_9}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$+\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3	$+4\lambda_7\,(\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$	$+4\lambda_7\,(\lambda_8)^*$

Quantity: λ_9

Terms 2 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda_9$
T2	$-3{g_1}^2\lambda_9$
T3	$+8\lambda_{12}^{(1)}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$

Quantity: λ_9

Terms 3 of 4

Column	Representative term structure
T1	$-4\lambda_{12}^{(A)}\lambda_9$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\ 2}$
T3	$+2\lambda_{S_1}^{(1)}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{12}^{(A)}\lambda_9$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\ 2}$	$+2\lambda_{S_1}^{(1)}\lambda_9$

Quantity: λ_9

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_9\lambda_{S_2}^{(1)}$
T2	$+5{\lambda_{10}}^2$
T3	$+2\lambda_{10}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$	$+2\lambda_{10}\lambda_{11}$

$\beta^{(1)}_{\lambda_{10}}$

Quantity: λ_{10}

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu} = \beta^{(1)}_{\lambda_{10}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$
T2	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T3	$-9{g_2}^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$

Quantity: λ_{10}

Terms 2 of 3

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{10}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2}\right)$

Quantity: λ_{10}

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda^{(1)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$-\frac{\lambda^{(A)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$
T3	$+12\lambda_{10}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda^{(1)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$-\frac{\lambda^{(A)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$	$+12\lambda_{10}\lambda^{(1)}_{S_2}$